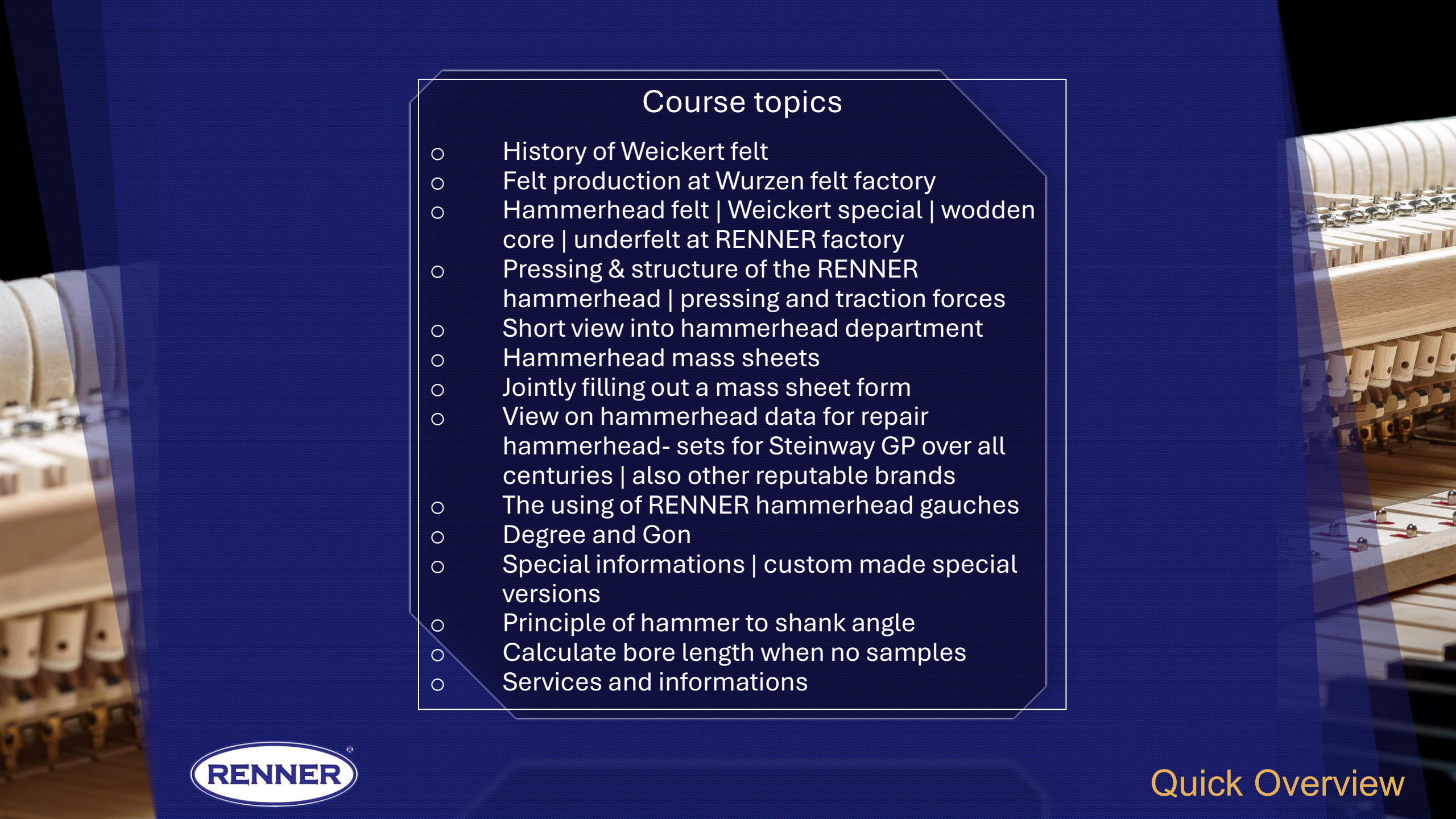
RENNER ACADEMY 2025

The production of the RENNER hammerhead and its selection for repair

Date: 04.10.2025

Creator: Jens- Sören Kruse | Klavierbaumeister | © RENNER ACADEMY





Course topics

- History of Weickert felt
- Felt production at Wurzen felt factory
- Hammerhead felt | Weickert special | wodden core | underfelt at RENNER factory
- Pressing & structure of the RENNER hammerhead | pressing and traction forces
- Short view into hammerhead department
- Hammerhead mass sheets
- Jointly filling out a mass sheet form
- View on hammerhead data for repair hammerhead- sets for Steinway GP over all centuries | also other reputable brands
- The using of RENNER hammerhead gauches
- Degree and Gon
- Special informations | custom made special versions
- Principle of hammer to shank angle
- Calculate bore length when no samples
- Services and informations

RENNER

Quick Overview

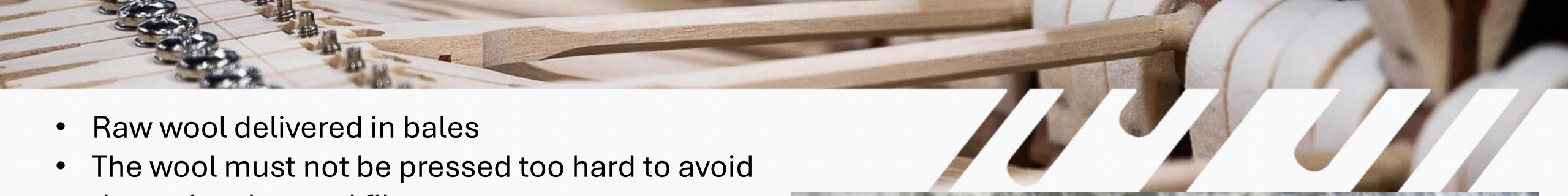


- Foundet in 1783 by J.D. Weickert in the town Leipzig in Saxony
- In 1847 the first hammerhead felt was produced in new factory Wurzen, that could be used like today
- Already a suplplier to many well known manufactures



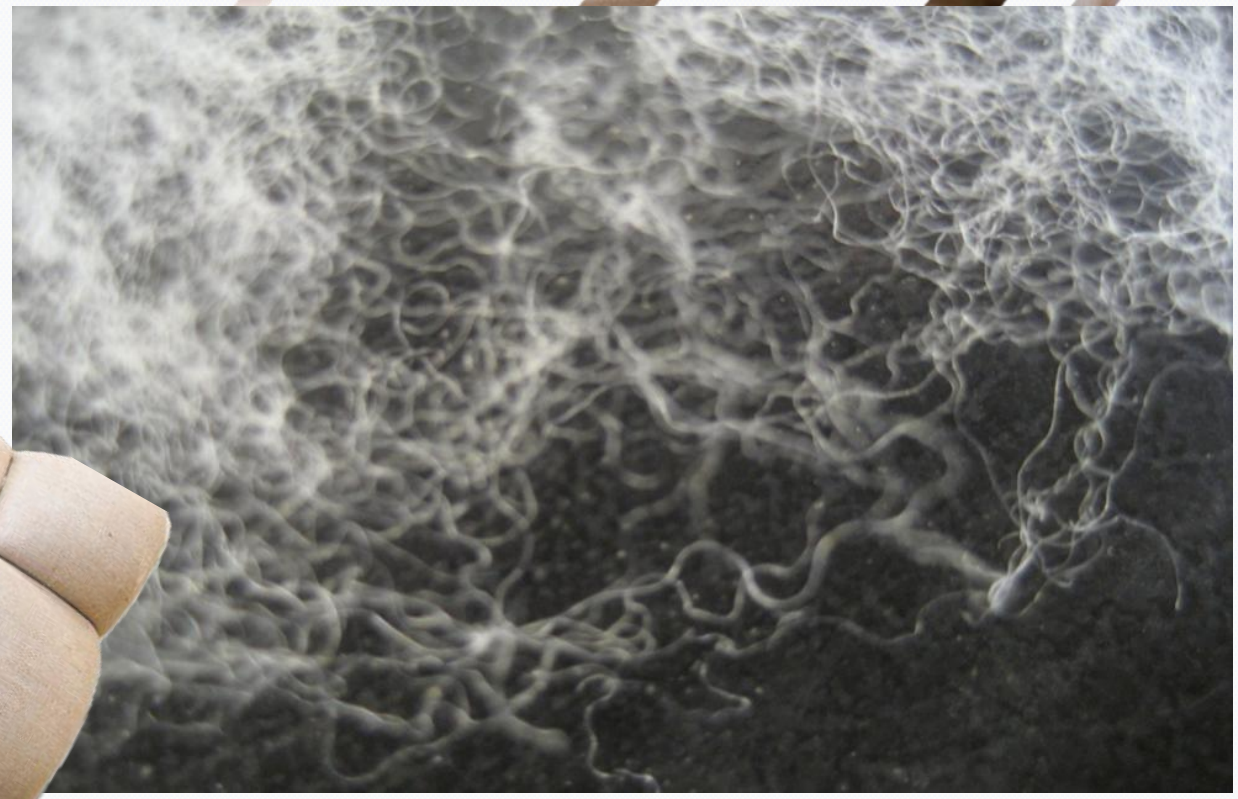
- Today owned by a felt manufacturer from Braunschweig/Germany and Toronto/Canada
- With the knowledge of the old employees, the hammerhead felt for all RENNER hammerheads is now produced in the modernized location in Wurzen

FFW Wurzen – Weickert and Wurzen felt today



- Raw wool delivered in bales
- The wool must not be pressed too hard to avoid damaging the wool fibres
- This increases transport costs but also the quality



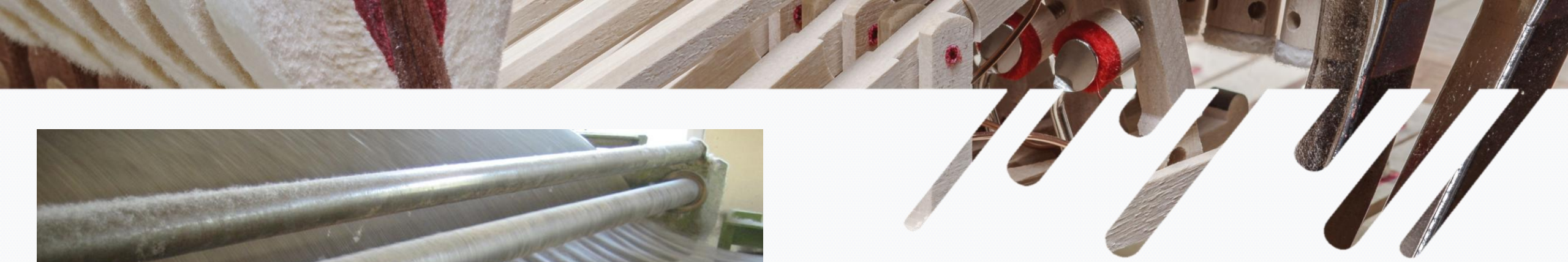


- If the wool fibres are strongly curled, this is an advantage for fulling during the felt production



- The blending and preparation of the wool fibers for RENNER hammerheads is a well-kept secret.
- Picture shows the loosening and the mixing process

Blending and mixing of wool fibres for RENNER hammerhead







- Combing the wool fibers into flow soft fleeces.
- Untangle and aligned fibers parallel to each other.
- The technical term for that is carding.



- Wool fleeces are layered on top of each other and united into a felt board





Felting needs:

- Steam
- Pressure
- Movement
- Know how

- Felt boards are compressed until the fibers can no longer be pushed together
- Corresponds to maximum elasticity
- Maximum natural walking hardness is 0.56 SHORE hardness
- Post-processing is done by compressing the felt boards with hammers and felt stones





- Re-pressing to the desired shore hardness and specified dimensions to the final size
- Without sanding and without reinforced pressing.
- This results in the greatest possible elasticity in the felt board
- This is recognizable by a smooth surface when bending

- Felt boards on stock at the Renner factory in Meuselwitz
- Humidity controlled
- Moth monitoring

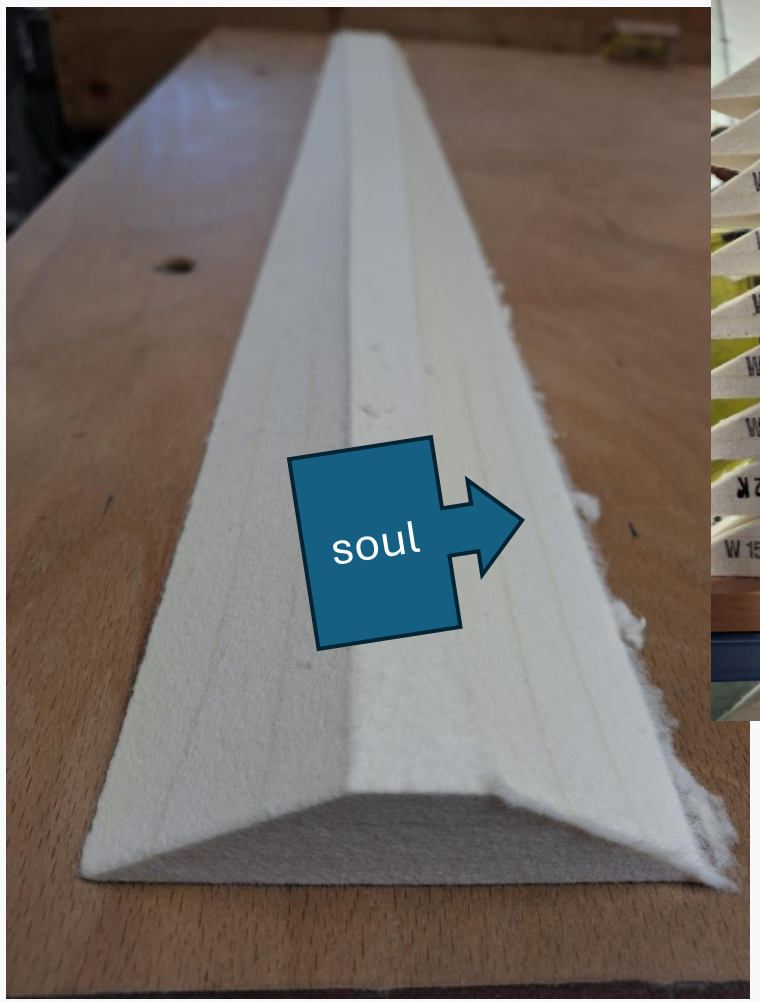


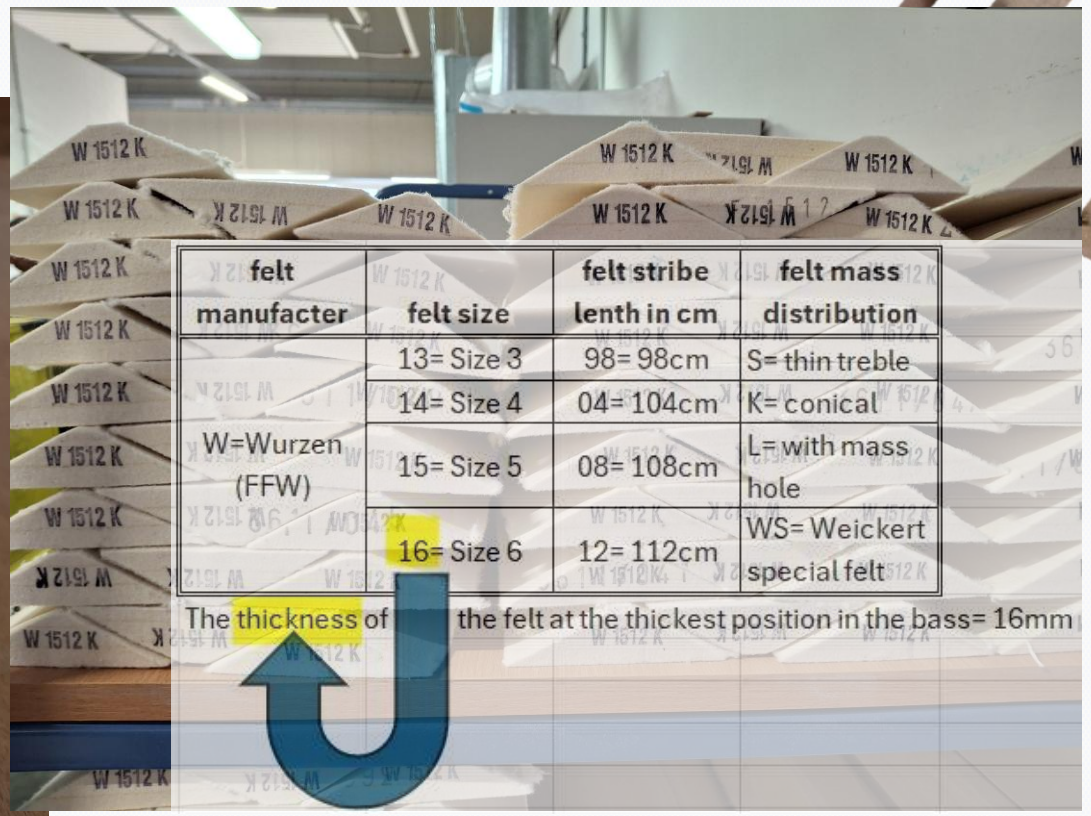


- Conical shape of the felt board
- Renner refers to well over 30 different felt boards from Wurzen felt factory, more than all

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Felt boards in humidity-controlled storage at RENNER factory





felt manufacter	felt size	felt stribe lenth in cm	felt mass distribution
W=Wurzen (FFW)	13= Size 3	98= 98cm	S= thin treble
	14= Size 4	04= 104cm	K= conical
	15= Size 5	08= 108cm	L= with mass hole
	16= Size 6	12= 112cm	WS= Weickert special felt

The thickness of the felt at the thickest position in the bass= 16mm

What is „Weickert special“ felt?

- It´s a spezial higher quality felt
- Only produced for RENNER
- Includes more wool grease
- This results in reduced swelling behaviour
- This results in better tracton behaviour
- This results in more stability of voicing and more directly tone

- Felt plates are cut into equal top felt stripes by a robot and signed with internal information for the production
- Some felt boards are delivered with 1, or 2 “souls”
- The soul is a different coloured layer of felt, mostly at the begin of bass and a mark of Wurzen felt factory
- No sign for different qualitys



Hornbeam

hard wood with a specific weight of 800-860 kg/m^3

Maple

medium-hard wood with a specific weight of 530-790 kg/m^3

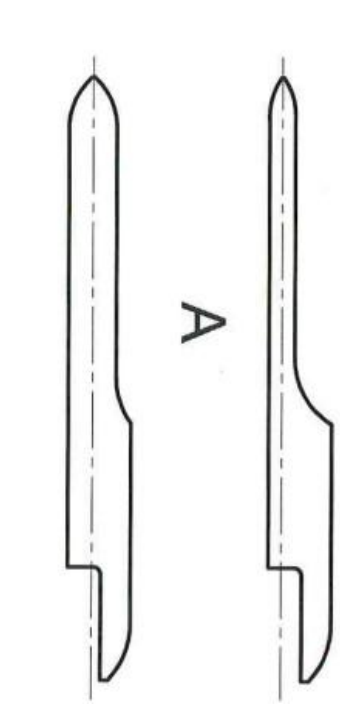
Walnut

medium-hard wood with a specific weight of 590-680 kg/m^3

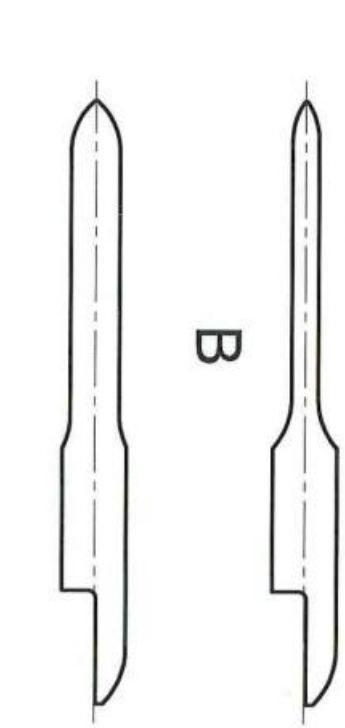
Mahogany (Kotibé)

medium-hard wood with a specific weight of 490-750 kg/m^3

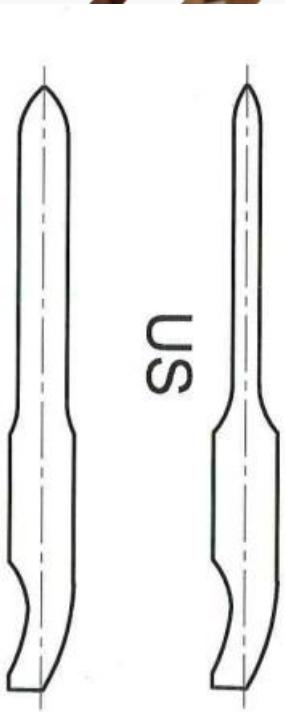




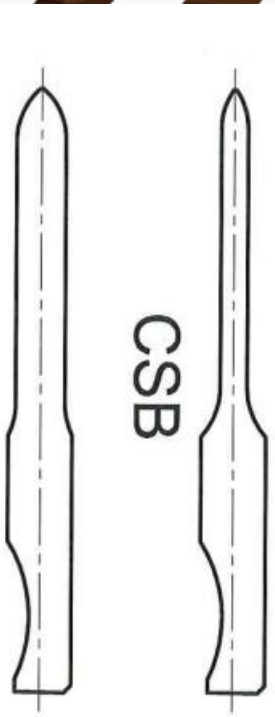
- Asymmetric form
- Old Bechstein
- GP with thin treble



- Symmetric standard form
- Bösendorfer
- Förster
- Petrof
- Sauter
- Grotrian Steinweg



- Steinway old and new in repair
- Yamaha
- Kawai
- Blüthner
- Renner USA



- Piano Industry
- Steinway HH
- Bösendorfer new
- Schimmel new
- Bechstein new

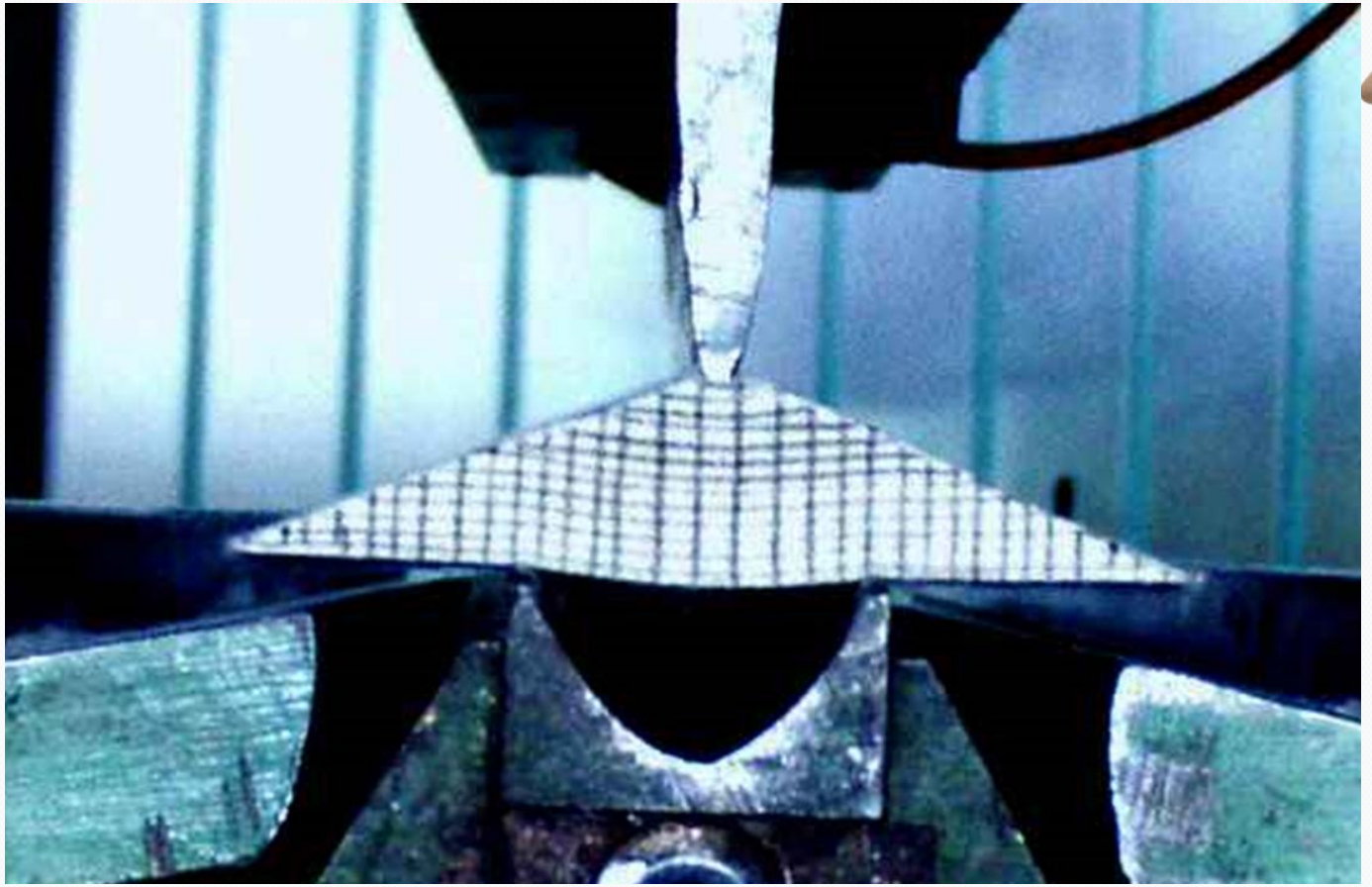
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Core moulding types

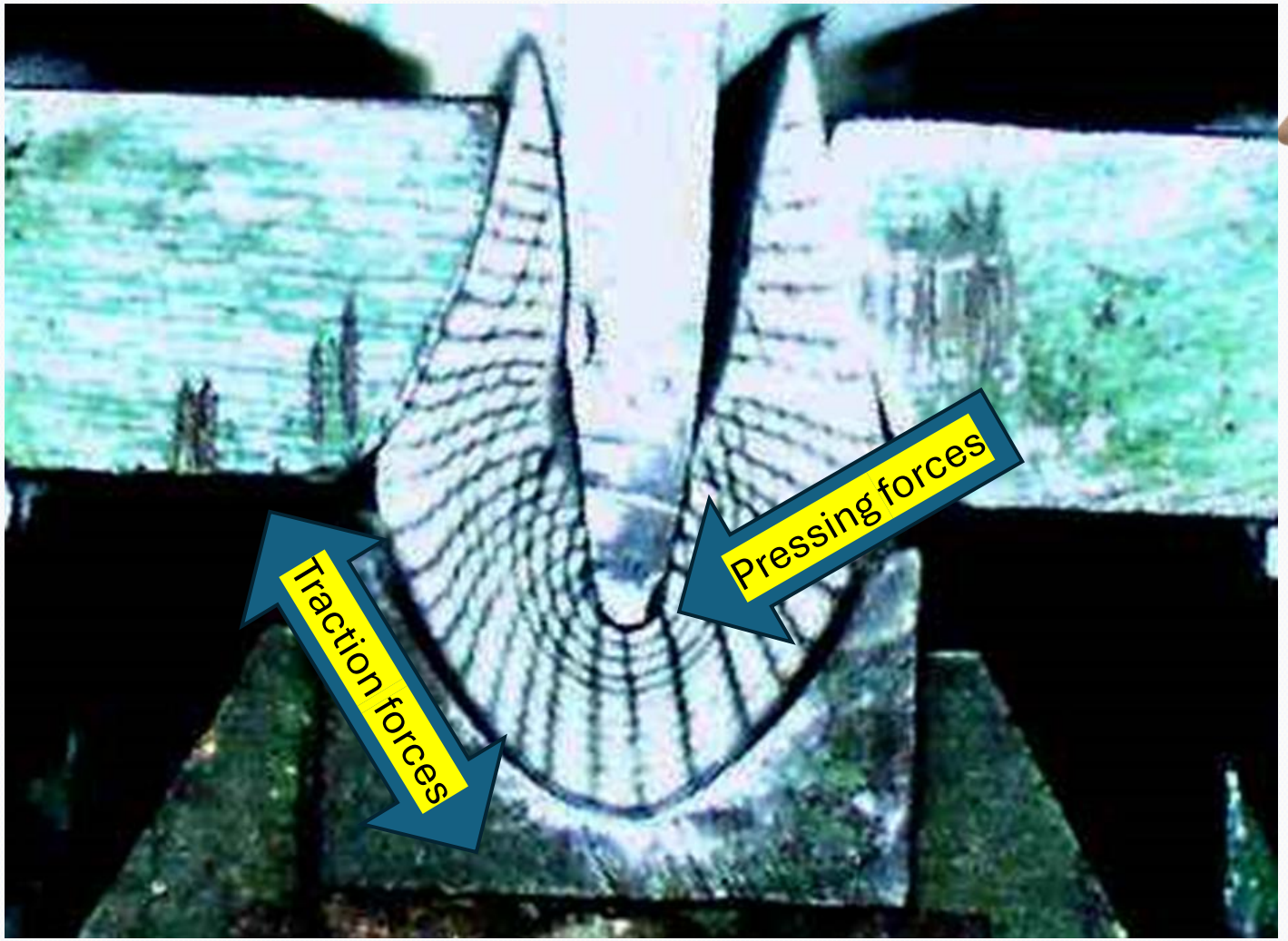


Underfelt colours standard program:

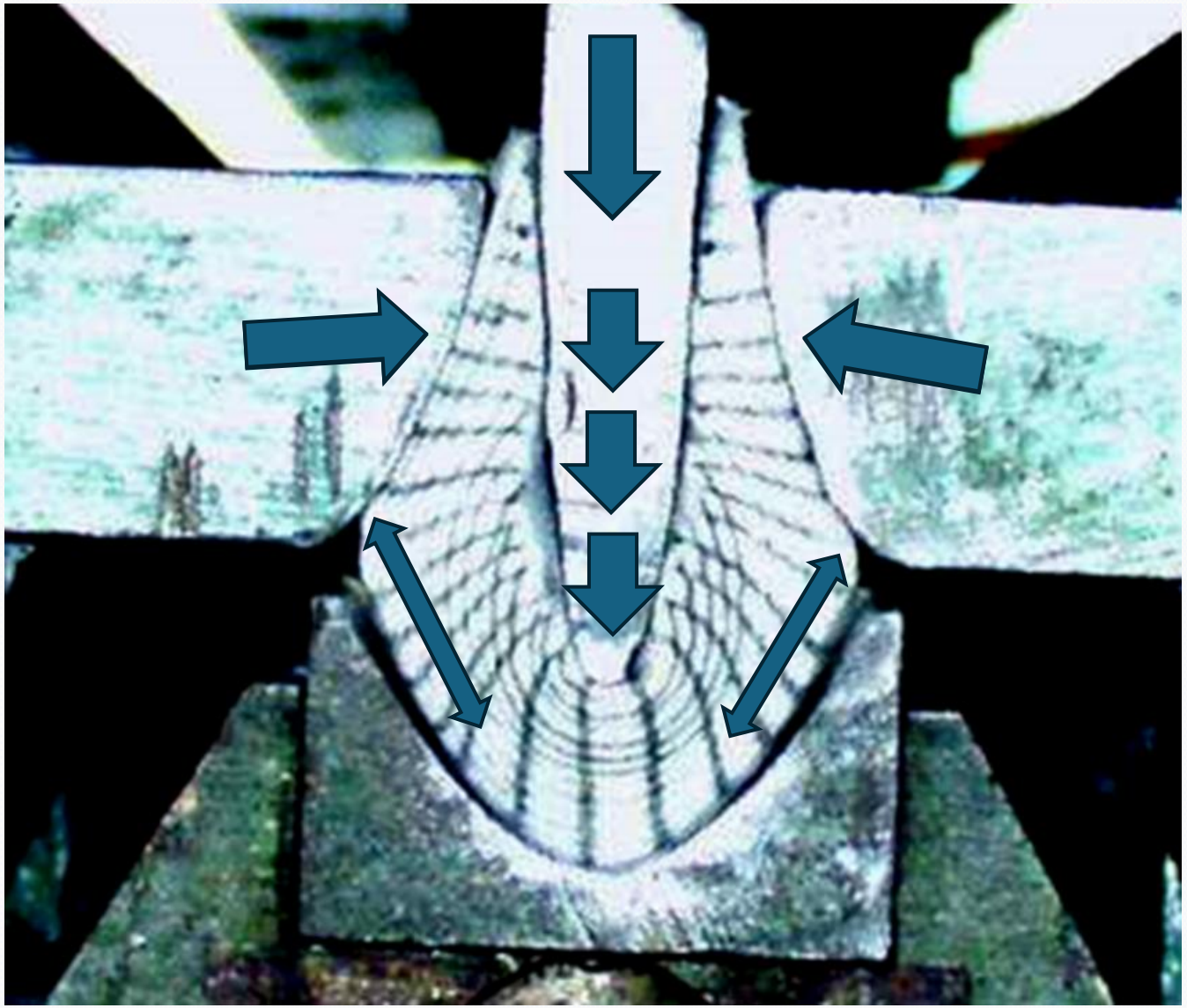
- red
- vine red
- orange
- yellow
- blue
- black



- Image shows the bass end of the felt strip before pressing.
- Characteristic curves on the felt.
- Observe changes during press process.

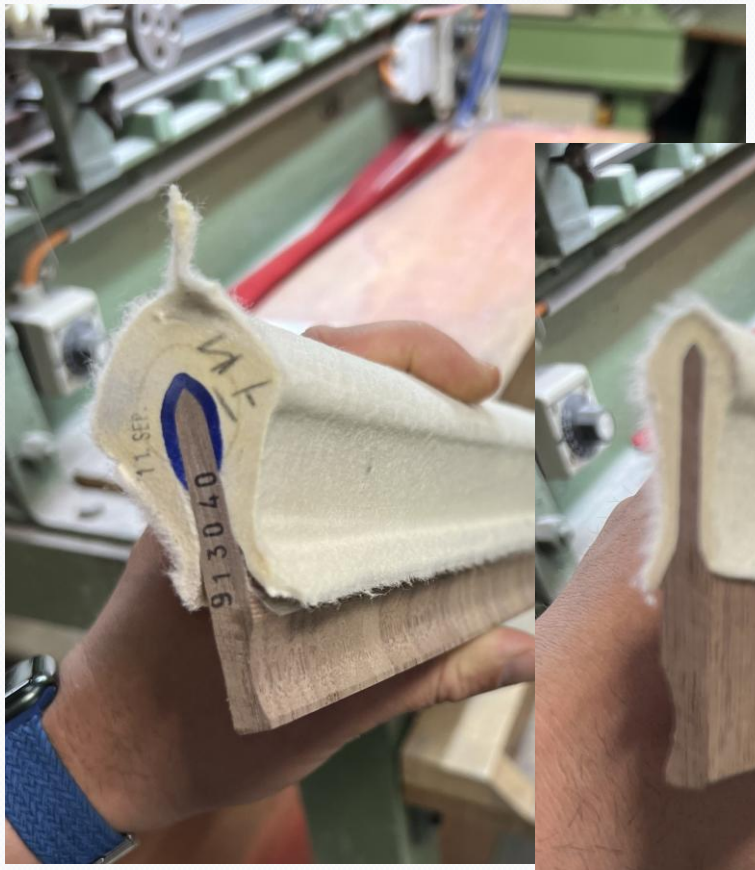


- Core is almost at the final position/ felt thickness on the peak.
- Counter-layers begin with the lateral closing of the felt around the core.
- Do you see how the characteristic curves have changed?
- Observation of the core lines, pressing and traction!



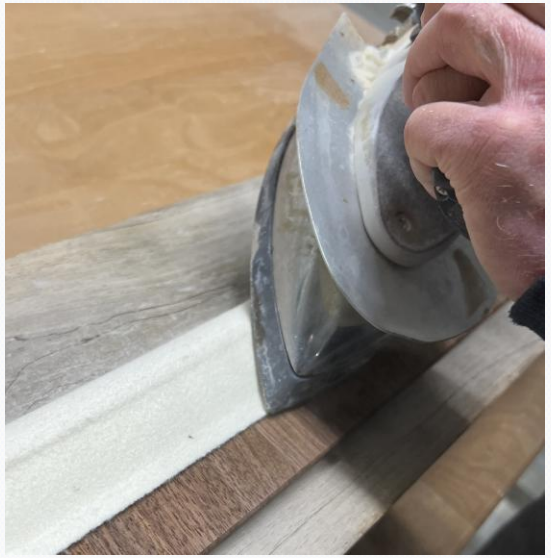


- Now the press is completely closed.
- Core was tensed up again after the side jaws were closed.
- This is done to additionally increase the tension on the outer felt layers.
- This is the basis of the work we want to carry out/demonstrate for tomorrow when we look for voicing process.



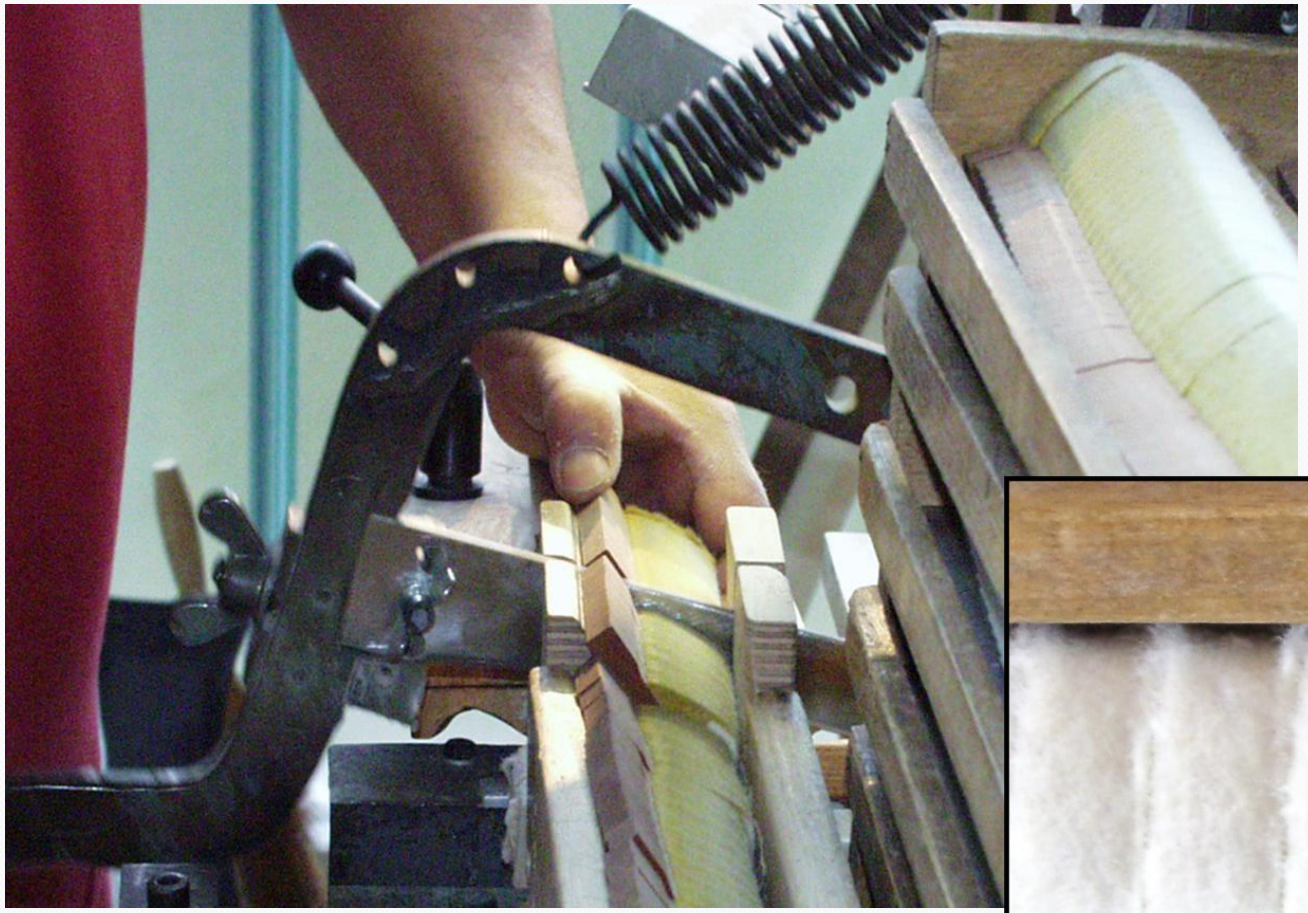


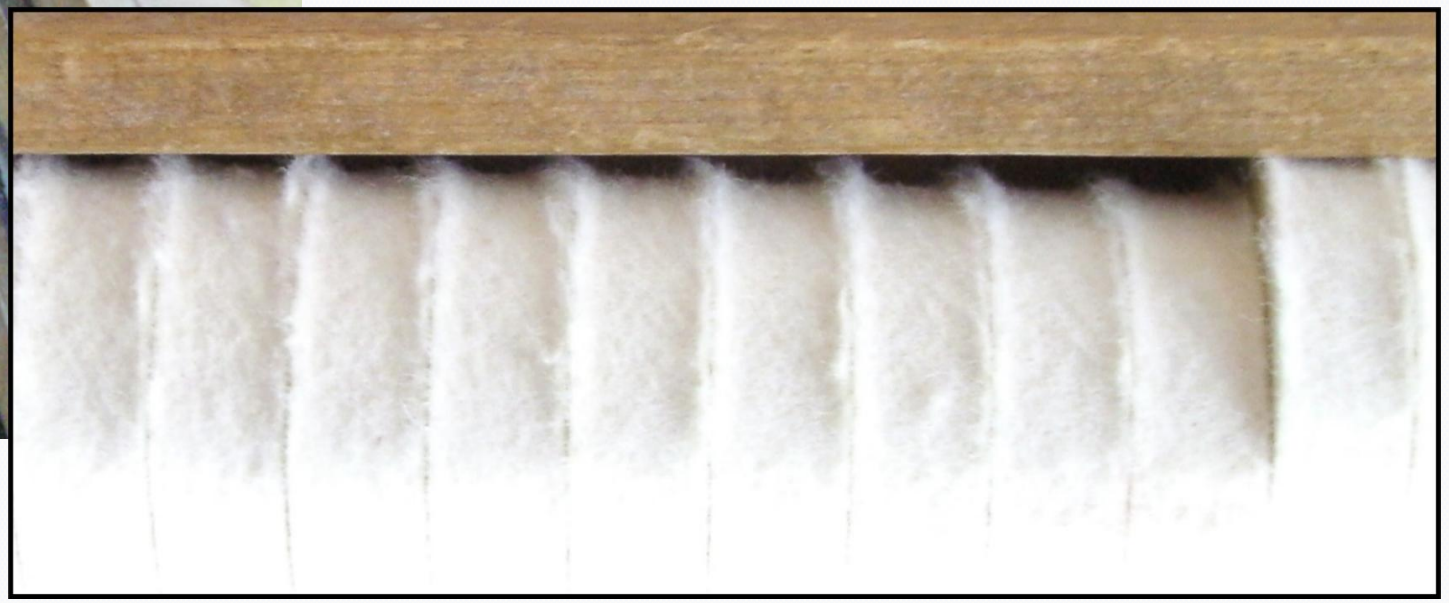
After gluing and pressing,
the hammerheads undergo a shaping process,
which we cannot show here for reasons
of maintaining trade and process secrets.



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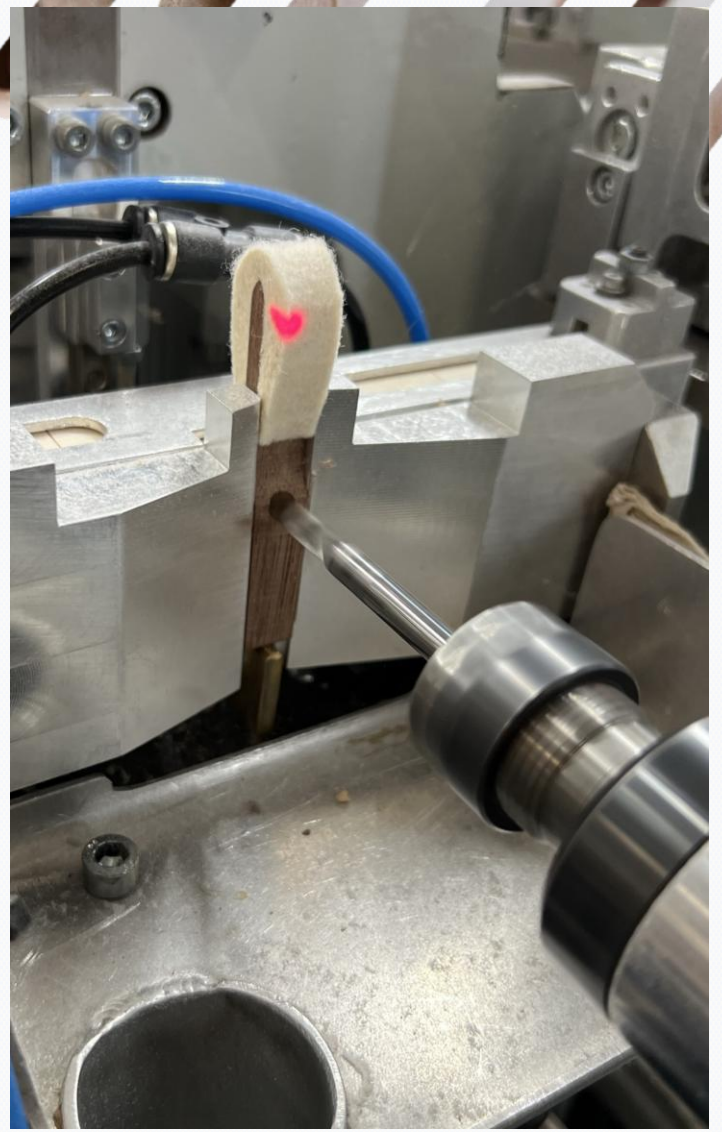
After pressing / shaping





Hammerhead surface after cutting





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Controlling | cleaning | drilling



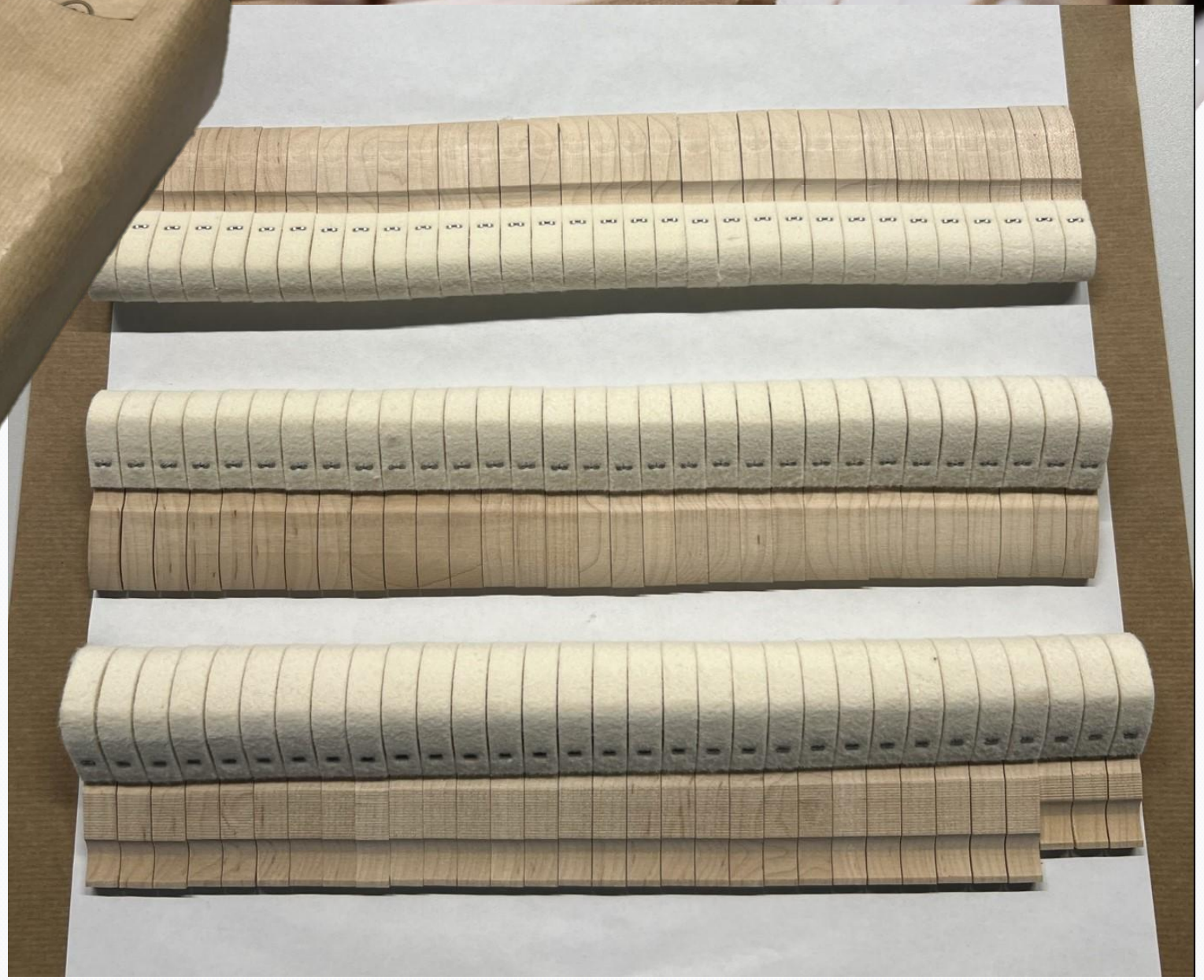




Install rivets




RENNER
Ahorn
Flügel-Hammerköpfe
Gr. 5 m. U. d. Teilung
ungebohrt
160/534







Measurement sheet for

GRAND PIANO HAMMERHEADS

RENNER®

Information concerning the grand piano

Type: _____ Year: _____ Size: _____ No. of sets which are to be produced: ____

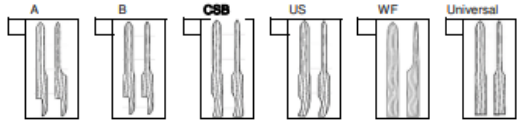
Information about the hammerheads

Size: ☐ Small / 4 ☐ Medium / 5 ☐ Large / 6 Weickert Special felt: ☐ Yes ☐ No

With underfelt: ☐ Yes ☐ No

Color: ☐ Red ☐ Wine-red ☐ Orange ☐ Yellow ☐ Blue ☐ Black

Form of the molding:



Wood: ☐ Hornbeam ☐ Kotibé ☐ Walnut ☐ Maple

Riveting: ☐ With rivet ☐ Without rivet

Additional work

Hammerhead-molding: tapered ☐ Yes ☐ No

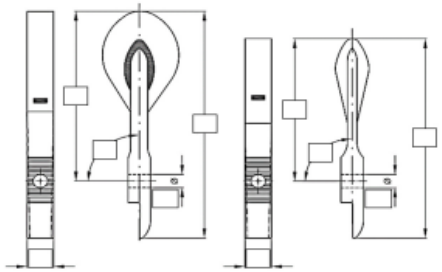
rounding ☐ Yes ☐ No

Felt cut in the bass: ☐ 1x ☐ 2x

Hammer boring: ☐ Yes ☐ No

Renner Standard: ☐

According to specifications: ☐
(To be measured with Renner hammerheads angel measurer)



Drilling inclinations	Bass												Treble											
Pieces																								
Degree																								

Others: _____

Address for the invoice:	Address for the delivery:

Date: _____ Signature: _____

Louis Renner GmbH
Max-Planck-Straße 18 - 24
D - 71116 Gärtingen
Telefon: + 49 (0)7034 / 64 50 - 0
Telefax: + 49 (0)7034 / 64 50 - 68
E-M@il: info@louis-renner.de



Measurement sheet for

PIANO HAMMERHEADS

RENNER®

Information concerning the grand piano

Type: _____ Year: _____ Size: _____ No. of sets which are to be produced: ____

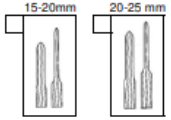
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Size: ☐ Small / 3 ☐ Medium / 4 ☐ Large / 5 Weickert Special felt: ☐ Yes ☐ No

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Color: ☐ Red ☐ Wine-red ☐ Orange ☐ Yellow ☐ Blue ☐ Black

Form of the molding:



Wood: ☐ Hornbeam ☐ Kotibé ☐ Walnut ☐ Maple

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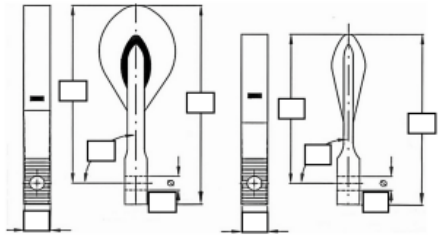
Additional work

Felt cut in the bass: ☐ 1x ☐ 2x

Hammer boring: ☐ Yes ☐ No

Renner Standard: ☐

According to specifications: ☐
(To be measured with Renner hammerheads angel measurer)



Drilling inclinations	Bass												Treble											
Pieces																								
Degree																								

Others: _____

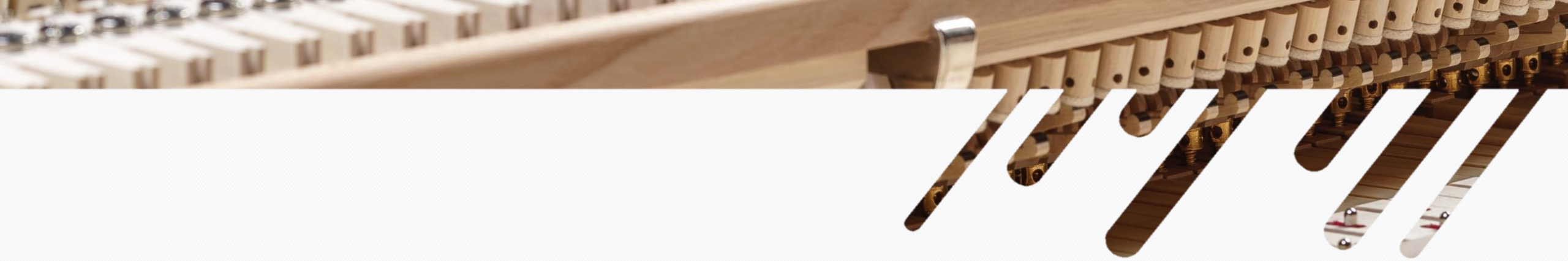
Address for the invoice:	Address for the delivery:

Date: _____ Signature: _____

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Max-Planck-Straße 18 - 24
D - 71116 Gärtingen
Telefon: + 49 (0)7034 / 64 50 - 0
Telefax: + 49 (0)7034 / 64 50 - 68
E-M@il: info@louis-renner.de

RENNER®

The RENNER hammerhead meas sheet for GP and UP





Measurement sheet for

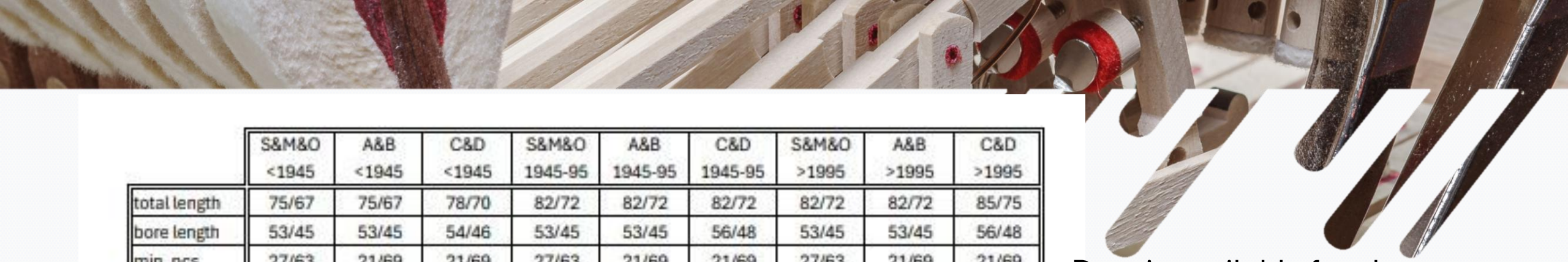
GRAND PIANO HAMMERHEADS

RENNER®

Information concerning the grand piano #535100

Type: Steinway & Sons Year: 1996 Size: M 170 No. of sets which are to be produced: 1

- For brand instruments, these details are already sufficient for RENNER to provide you with a suitable, drilled hammer head set, ready to install.



	S&M&O <1945	A&B <1945	C&D <1945	S&M&O 1945-95	A&B 1945-95	C&D 1945-95	S&M&O >1995	A&B >1995	C&D >1995
total length	75/67	75/67	78/70	82/72	82/72	82/72	82/72	82/72	85/75
bore length	53/45	53/45	54/46	53/45	53/45	56/48	53/45	53/45	56/48
min. pcs	27/63	21/69	21/69	27/63	21/69	21/69	27/63	21/69	21/69
shenk angle	90°	90°	90°	90°	90°	90°	90°	90°	90°
size	4-sD-I	4-sD-I	4-sD-I	5	5	5	5	5	6
form	US	US	US	US	US	US	US	US	US
cutting width	11,2	11,2	11,2	11,2	11,2	12	11,2	11,2	12
wood	walnut	walnut	walnut	mahogany	mahogany	mahogany	maple	maple	maple
underfelt	yellow	yellow	yellow	vine red	vine red	vine red	vine red	vine red	vine red
with rivets	yes	yes	yes	yes	yes	yes	yes	yes	yes
bore ø	5,4	5,4	5,4	5,4	5,4	5,4	5,4	5,4	5,4
Renner Art. Nr. old	519160426	519160420	519160423	519160507	519160505	519160630	519160534	519160533	519160611
Renner Art. Nr. new	43-0001752	43-0001752	43-0001752	43-0001752	43-0001752	43-0001752	43-0001752	43-0001752	43-0001752
Variant	V0002077	V0002068	V0002072	V0002076	V0002078	V0001899	V0001945	V0001947	V0001908

	Bass					Treble										
Gon	6	7	8	9	10	110	109	108	107	106	105	104	103	102	101	100
S+M+O					1-26	27-30	31-33	34-36	37-39	40-42	43-45	46-48	49-51	52-54	55-88	
A+B				1-20			21-26	27-32	33-35	36-38	39-41	42-44	45-47	48-51	52-68	69-88
C			1-8	9-20			21-26	27-38	39-44	45-47	48-50	51-53	54-59	60-65	66-88	
D	1-6	7-12	13-20					21-26	27-32	33-36	37-40	41-44	45-48	49-63	64-88	

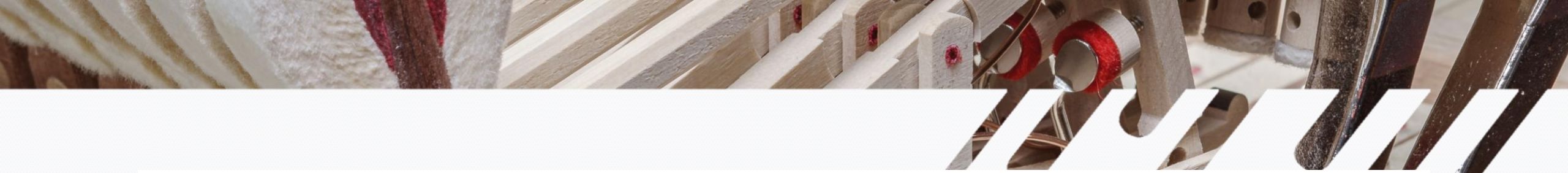
Data is available for almost all reputable brands for example:

- Bechstein
- Blüthner
- Bösendorfer
- Schimmel
- Sauter
- Steingraeber
- Seiler
- Förster
- Yamaha
- KAWAI
- Petrof

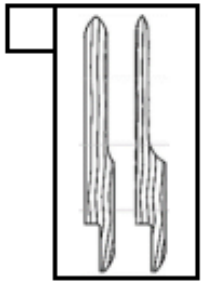
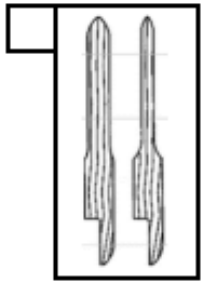
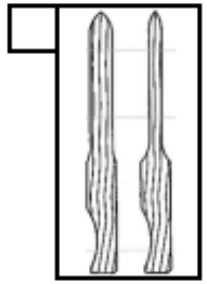
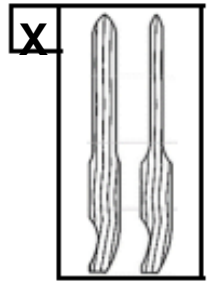
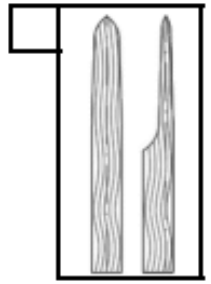
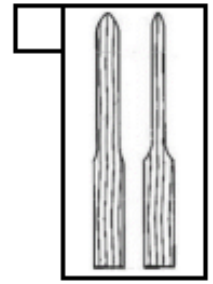
etc.

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Data sample RENNER standard program for Steinway repairs



Information about the hammerheads

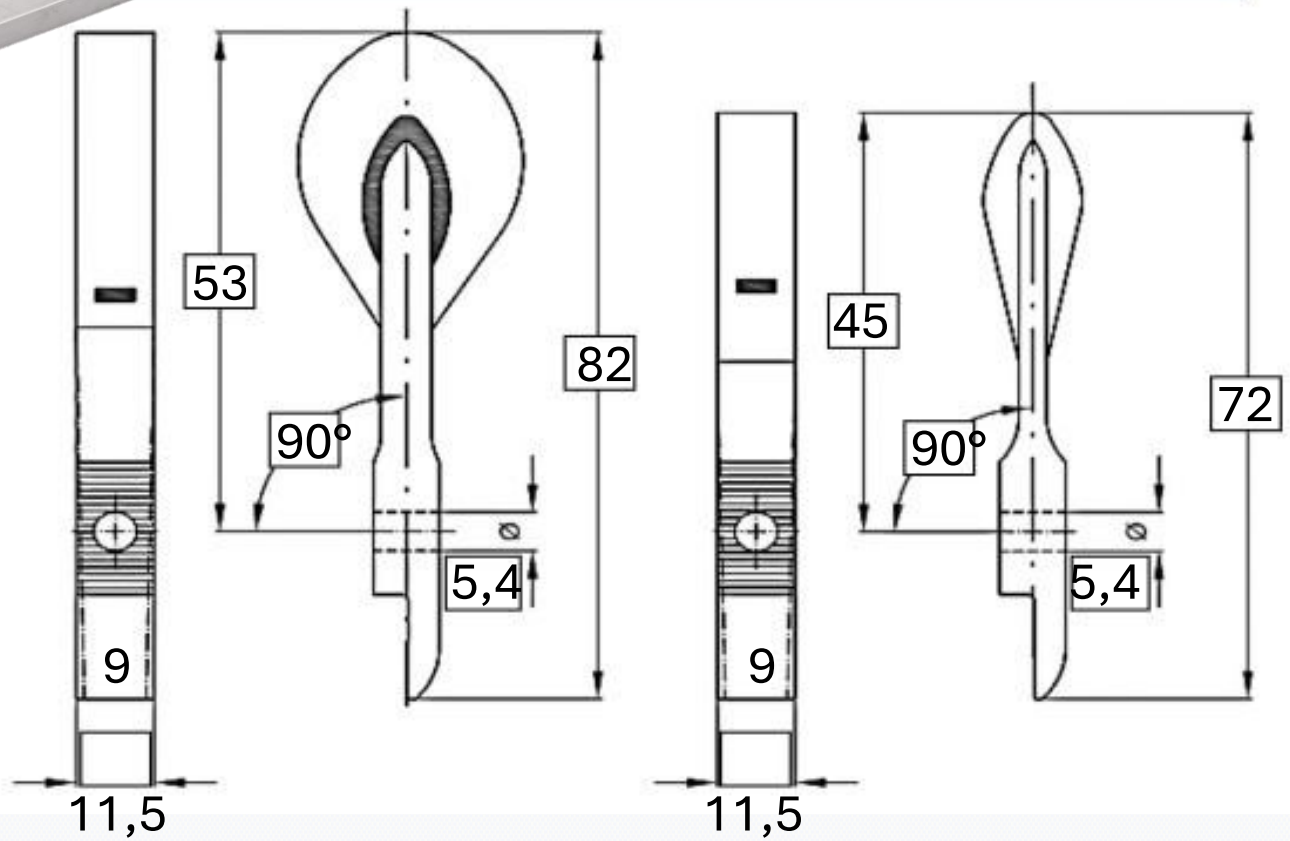
Size:	☐	Small / 4	☒	Medium / 5	☐	Large / 6	Weickert Special felt:	☐	Yes	☒	No
With underfelt:	☒	Yes	☐	No							
Color:	☐	Red	☒	Wine-red	☐	Orange	☐	Yellow	☐	Blue	☐ Black
Form of the molding:	☐	A 	☐	B 	☐	**CSB** 	☒	US 	☐	WF 	☐ Universal 
Wood:	☐	Hornbeam			☐	Kotibé			☐	Walnut	☒ Maple
Riveting:	☒	With rivet			☐	Without rivet					



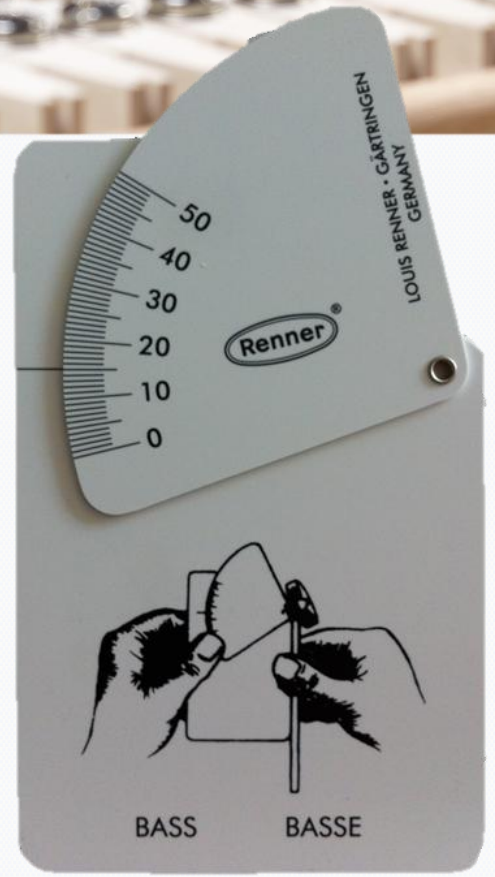
Additional work

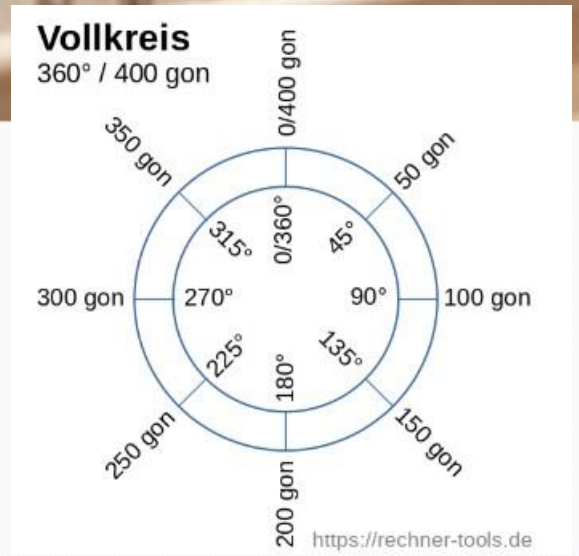
Hammerhead-molding:	tapered	[X] Yes	[ ] No
	rounding	[X] Yes	[ ] No
Felt cut in the bass:		[ ] 1x	[ ] 2x
Hammer boring:		[X] Yes	[ ] No
Renner Standard:		[X]	
According to specifications:		[X]	

(To be measured with Renner hammerheads angel measurer)



Renner Standard means: bore length = total length -24mm GP (or after model informations RENNER has) | bore ø 5,4mm | hammer to shenk angle 90° | drilling angles after RENNER informations.





- Drilling angles in Gon = 100 parts at a right angle

- Upright piano normal values: bass 30 Gon- 0 Gon and treble 125 Gon- 0 Gon
- Grand piano normal values: bass 25 Gon- 0 Gon and treble 125 Gon- 0 Gon

Drilling inclinations	Bass					Treble										
Pieces			1-26			27-30	31-33	34-36	37-39	40-42	43-45	46-48	49-51	52-54	55-88	
~~Degree~~ Gon			10			110	109	108	107	106	105	104	103	102	101	

Others: Minimum number of hammers bass = 27pcs and treble = 63pcs



Technical specifications

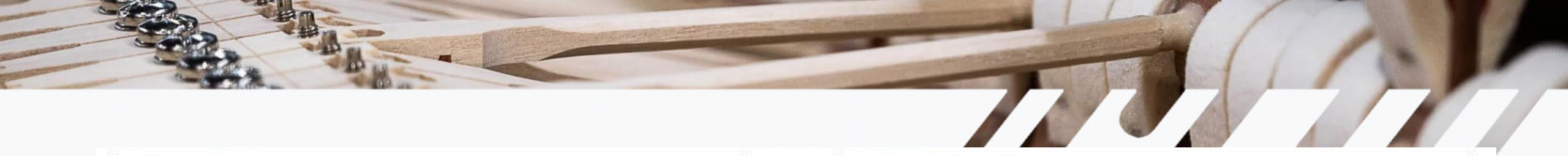
TL	BL	Pcs.	Size	Form	Cut. Width	Wood	U - felt	Art. Nr.	fitting for / optional version
Grand hammerheads with Weickert-Spezial premium felt									
75 / 67		21 / 69	4-sD-l	US	11,2	walnut	yellow	51 9 260 420	e.g. old Steinway&Sons A&B before 1950 bored like old number 519160421 (BL 53/45)
		27 / 63	4-sD-l	US	11,2	walnut	yellow	51 9 260 426	e.g. old Steinway&Sons S&O&M before 1950 bored like old number 519160427 (BL 53/45)
80 / 72		24 / 69	5	US	11,0	walnut	yellow	51 9 260 007	e.g. Yamaha C7
		26 / 65	5	US	11,0	walnut	wine red	51 9 260 550	e.g. Yamaha C3,5,6
80 / 75		30 / 68	4-sD-l	US	10,7	hornbeam	without	51 9 260 438	lighter version
		28 / 64	4-sD	A	11,0	mahogany	without	51 9 160 516	e.g. Bechstein A&B&L&M&S&V - softer version
		27 / 71	6	US	10,5 / 10,8	hornbeam	orange	51 9 260 600	NEW e.g. for Bösendorfer
81 / 75		28 / 64	4-sD-l	US	10,5 / 11,0	hornbeam	blue	51 9 260 439	e.g. Blüthner - lighter version
82 / 72		21 / 69	5	US	11,5	maple	wine red	51 9 260 533	e.g. newer Steinway&Sons A&B since 1995
		27 / 63	5	US	11,5	maple	wine red	51 9 260 534	e.g. newer Steinway&Sons S&O&M since 1995
82 / 73		30 / 64	4	US	11,0	walnut	yellow	51 9 260 002	e.g. Yamaha G2/G3
83 / 75		29 / 63	5	US	11,0	walnut	yellow	51 9 260 003	e.g. Yamaha C3,5,6
		30 / 62	4	B	10,7 / 10,8	hornbeam	red	51 9 260 447	
		30 / 68	4-sD-l	US	10,7	hornbeam	without	51 9 260 448	lighter version
		29 / 66	5	US	10,5 / 10,8	hornbeam	red	51 9 260 777	NEW e.g. for Grotrian-Steinweg
83 / 78		27 / 71	5	CSB	10,5 / 10,8	hornbeam	orange	51 9 260 501	e.g. Bösendorfer - universal
86 / 76		20 / 71	4-sD	A	11,0	mahogany	without	51 9 160 514	e.g. Bechstein C&E - softer version
88 / 80		28 / 71	5	B	10,7 / 10,7	hornbeam	orange	51 9 260 514	e.g. old Bösendorfer

description

TL = Total length
BL = Bored length
Pcs. = Minimum pieces of bass/treble hammers
Form = moulding form
Cut. width = cutting width bass/treble hammers
Wood = moulding wood
U-felt = underfelt
size:
RENNER offers their hammerheads in 3 different sizes (3-5 on upright hammers, 4-6 on grand hammers).
With these sizes we try to visualize the differences between "light", "middle" and "heavy".
We are sorry that we can not give you an exact hammer weight per set since there are too many factors which do influence the absolute weight per hammer.
Some of these are the cutting width, the tail shape of the hammer moulding, the rivets and last but not least the selection of the top cover felt.
Please notice the following additional details of the column "size":
sD = slim treble hammers (The volume of the treble hammers has been reduced. This is necessary when you have pianos where the distance to the cast plate is very tight, e.g. old Bechstein grands)
WS = Weickert-Spezial premium felt
l = light version (e.g. size "5, l" means, that the weight is between normal size 4 and 5).
w = softer version (as compared to the RENNER Standard version)
geb. = only available in already bored version, after the knoweldge of RENNER

RENNER®

Data for standard hammerheads and descriptions



Address for the invoice:

Your headquarter adress

Address for the delivery:

Your workshop adress

Date: 20.10.2025

Signature: Your sign

Louis Renner GmbH
Max-Planck-Straße 18 - 24
D - 71116 Gärtingen

Telefon: + 49 (0)7034 / 64 50 - 0
Telefax: + 49 (0)7034 / 64 50 - 68
E-M@il: info@louis-renner.de

Renner China

Tel. **0086-13676590202**
E-Mail **ozmei@outlook.com**

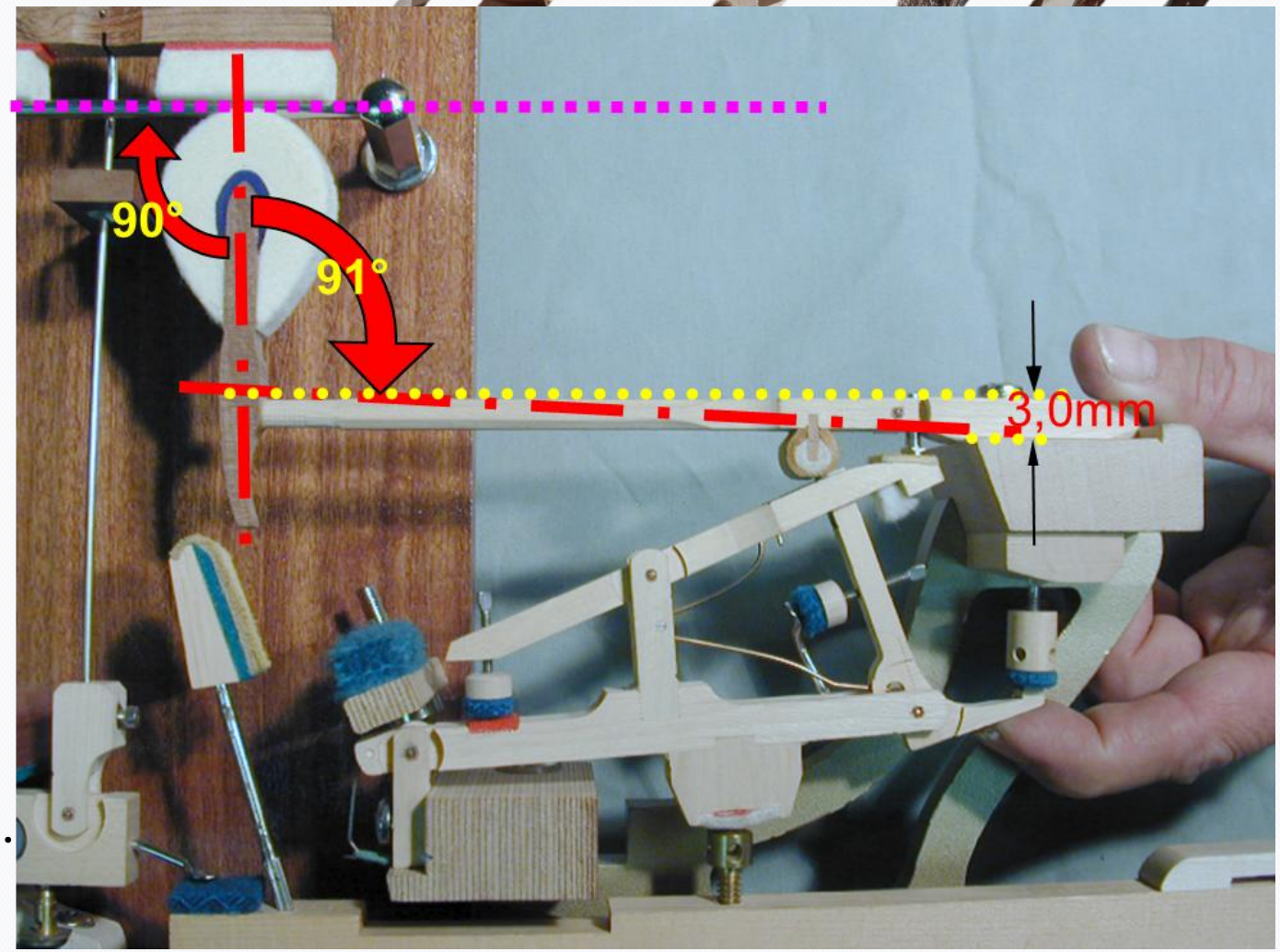
Ouzhimei (Shanghai) Art & Culture Co. Ltd
66 E. Guangfulin Rd., Songjiang District
Shanghai
201613
China

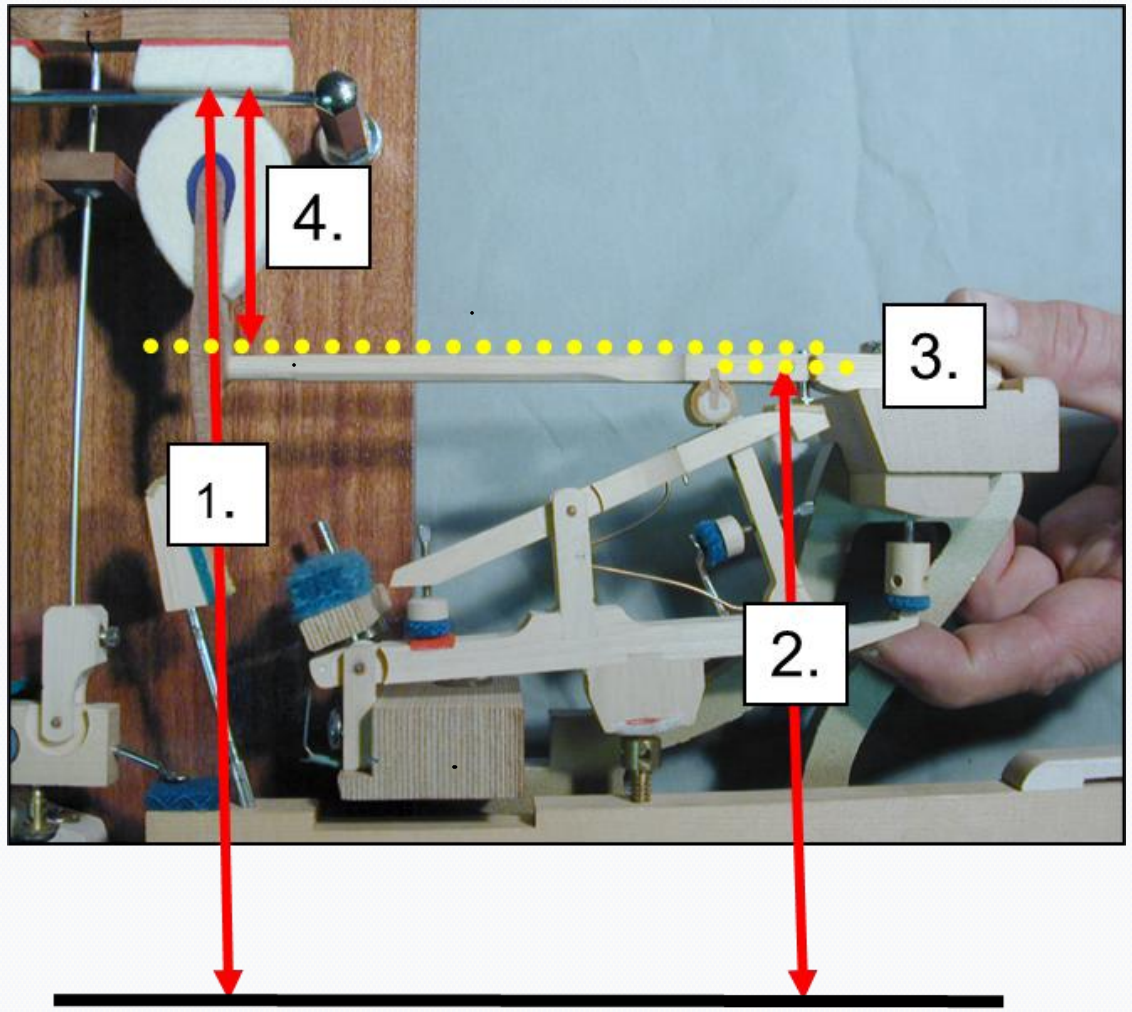
Principle:

- The hammer should hit at a right angle to the plane of the strings.
- The hammer shank should be able to make a 3 mm overhang.
- The effect is that due to the inertia of the material, the hammer excites the energy in the direction of the bridge more.
- If the angle becomes smaller than 90° when hit the string, the hammer gets wedged towards to the agrafe and the string can break and affecting the tone.

When does this happens mostly?

- After repairing old hammerheads, when to much felt is sanded from the hammerhead.
- Good argument to sell new hammers to your customer.





(1.) Distance from string plane to keybed
- (2.) Distance hammer shank axis (center pin) to keybed
= Difference between both values
- (3.) 3 mm (for overhang shank when hits string)
= (4.) drilling length

Calculation of boreing length



Measurement sheet for GRAND PIANO HAMMERHEADS

RENNER®

Information concerning the grand piano

Type: _____ Year: _____ Size: _____ No. of sets which are to be produced: ____

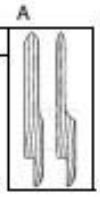
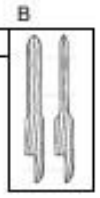
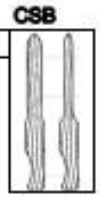
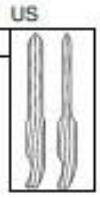
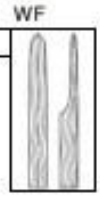
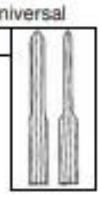
Information about the hammerheads

Size: ☐ Small / 4 ☐ Medium / 5 ☐ Large / 6 Weickert Special felt: ☐ Yes ☐ No

With underfelt: ☐ Yes ☐ No

Color: ☐ Red ☐ Wine-red ☐ Orange ☐ Yellow ☐ Blue ☐ Black

Form of the molding:

A	B	CSB	US	WF	Universal
☐ 	☐ 	☐ 	☐ 	☐ 	☐ 

Wood: ☐ Hornbeam ☐ Kotibé ☐ Walnut ☐ Maple

Riveting: ☐ With rivet ☐ Without rivet

Additional work

Hammerhead-molding: tapered ☐ Yes ☐ No

rounding ☐ Yes ☐ No

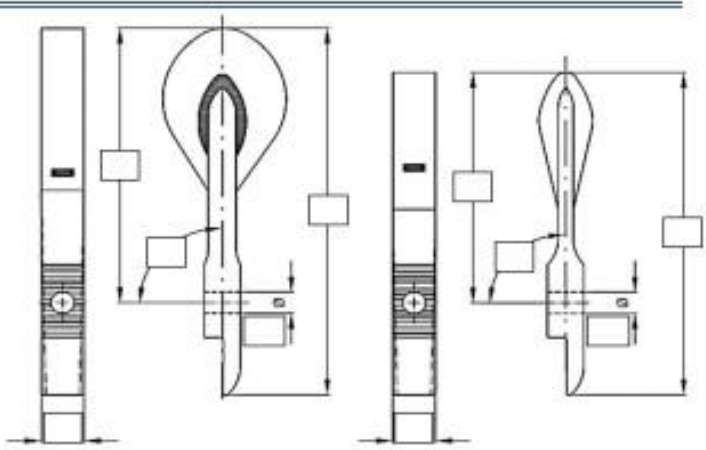
Felt cut in the bass: ☐ 1x ☐ 2x

Hammer boring: ☐ Yes ☐ No

Renner Standard: ☐

According to specifications: ☐

(To be measured with Renner hammerheads angel measurer)



Drilling inclinations	Bass					Treble												
Pieces																		
Degree																		

Others: _____

Address for the invoice:

Address for the delivery:

Date: _____ Signature: _____

Louis Renner GmbH
Max-Planck-Straße 18 - 24
D - 71116 Gärtingen

Telefon: + 49 (0)7034 / 64 50 - 0
Telefax: + 49 (0)7034 / 64 50 - 68
E-M@il: info@louis-renner.de

GP measurement sheet

Renner China

Tel. 0086-13676590202
E-Mail ozmei@outlook.com

Ouzhimei (Shanghai) Art & Culture Co. Ltd
66 E. Guangfulin Rd., Songjiang District
Shanghai
201613
China



Measurement sheet for PIANO HAMMERHEADS

RENNER®

Information concerning the grand piano

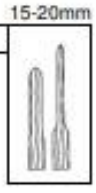
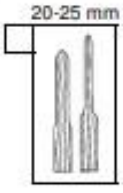
Type: _____ Year: _____ Size: _____ No. of sets which are to be produced: ____

Information about the hammerheads

Size: ☐ Small / 3 ☐ Medium / 4 ☐ Large / 5 Weickert Special felt: ☐ Yes ☐ No

With underfelt: ☐ Yes ☐ No

Color: ☐ Red ☐ Wine-red ☐ Orange ☐ Yellow ☐ Blue ☐ Black

Form of the molding: ☐ 15-20mm  ☐ 20-25 mm 

Wood: ☐ Hornbeam ☐ Kotibé ☐ Walnut ☐ Maple

Riveting: ☐ With rivet ☐ Without rivet

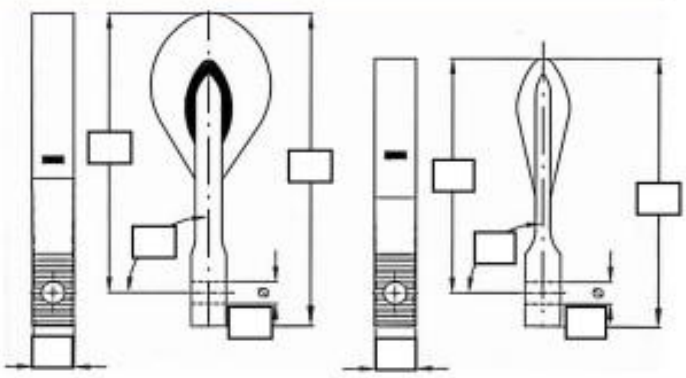
Additional work

Felt cut in the bass: ☐ 1x ☐ 2x

Hammer boring: ☐ Yes ☐ No

Renner Standard: ☐

According to specifications: ☐
(To be measured with Renner hammerheads angel measurer)



Drilling inclinations	Bass									Treble								
Pieces																		
Degree																		

Others: _____

Address for the invoice:	Address for the delivery:

Date: _____ Signature: _____

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D - 71116 Gärtingen

Telefon: + 49 (0)7034 / 64 50 - 0
Telefax: + 49 (0)7034 / 64 50 - 68
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UP measurement sheet

Renner China

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E-Mail ozmei@outlook.com

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66 E. Guangfulin Rd., Songjiang District
Shanghai
201613
China



If interess look at YouTube Wurzen Felt Channel

https://youtu.be/lA7lZXILLk4?si=KhoTi8K6txE7JSRi

Thank you for your interest

Let the Perfect Hammer Sound!

RENNER®

